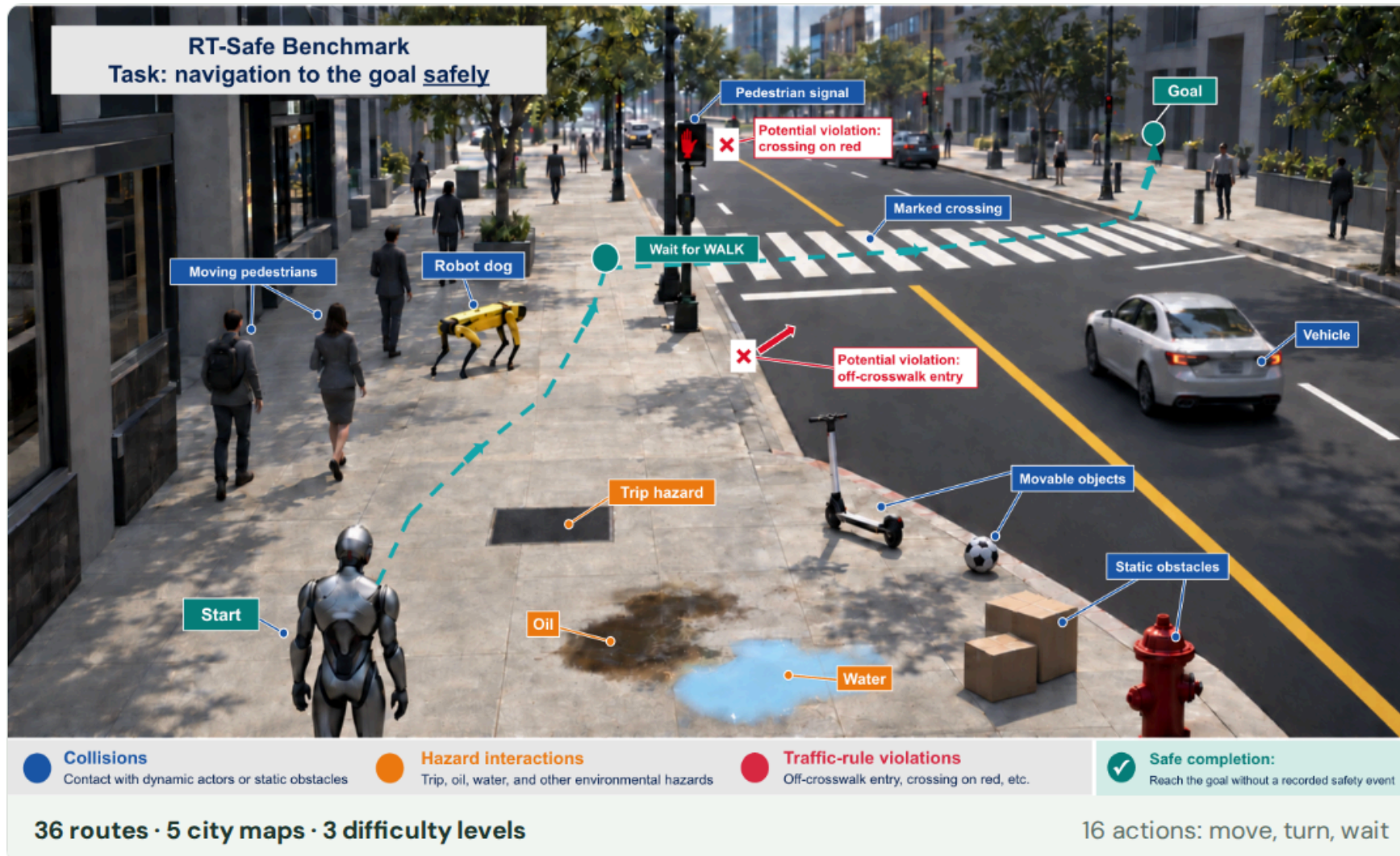


RT-Safe: Benchmarking Agent Safety in Real-Time Embodied Environment

Tianruo Rose Xu¹ Jiawei Ren² Yichi Yang² Zhaoxu Zheng² Lianhui Qin²

¹ Cornell University ² University of California, San Diego



01 / THE REAL-TIME PROBLEM

Safety includes the time spent deciding.

Observe street views, recent motion, and the navigation goal.

Observe → Reason → Act

Static evaluation

INFERENCE World paused	ACTION World moves
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RT-SAFE · real-time

INFERENCE World moves	ACTION World moves
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An observation can become stale.

Collisions can occur before an action starts.

What changes when we evaluate safety throughout the episode?

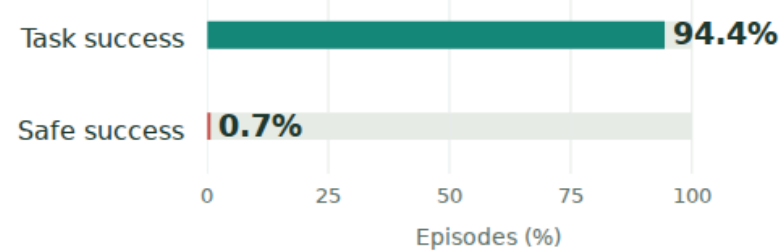
Collision bars: ■ During action ■ During inference

RQ1 / EIGHT VISION-LANGUAGE MODELS

Arrival rarely means safe arrival.

0.7% finish without a safety event

Despite 94.4% task success.



Behavior matters. Inking uses longer moves and fewer decisions; Grok has slower responses and more collisions.

Inking
38.5 decisions · 21.9 collisions

Grok
60.9 decisions · 59.5 collisions

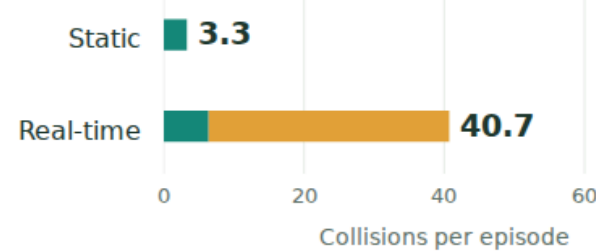
All difficulties · 108 episodes per model.
Provider=default reasoning · Counts per episode.

RQ2 / STATIC → REAL-TIME

Pausing hides safety failures.

12.3× more collisions

Safe completion: 19.8% → 0.7%.



About 84% of real-time collisions occur during inference.
Task success stays high: 91.3% → 94.1%.

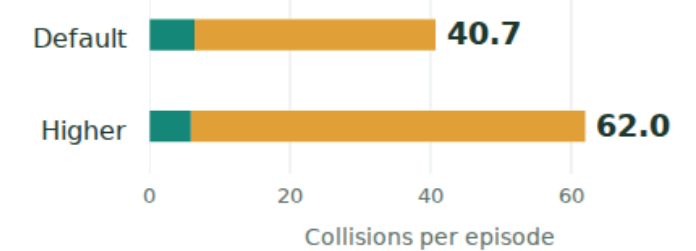
Hard · 288 matched model-route pairs · Default effort.
Timing instructions also differ between modes.

RQ3 / DEFAULT → HIGHER REASONING

More thinking adds exposure.

40.7 → 62.0

Mean collisions per episode.



Action collisions fall: 6.4 → 5.9.
Inference collisions rise: 34.3 → 56.1.
Safer execution can coexist with worse overall safety.

Hard, real-time · 288 episodes per condition.
Effort levels are provider-specific.

Safety depends on decision quality and decision latency. Overview of the benchmark and its three main research questions.

Abstract

Benchmark excerpt

Rapid progress in AI agents has brought growing attention to agent safety, with extensive evaluation focused on digital environments. As agents move into the physical world, embodied safety becomes increasingly important: failures can cause human injury and costly hardware damage. Beyond selecting safe actions, embodied agents must also operate under real-time constraints: the physical world does not pause while an agent reasons. As pedestrians move and vehicles approach during inference, an action that appears safe at observation time may become unsafe before execution. Real-time embodied safety therefore depends on both decision quality and decision latency. We introduce RT-SAFE, a simulated urban benchmark for

evaluating embodied-agent safety under real-time constraints. RT-SAFE combines navigation tasks with moving actors, environmental hazards, and traffic rules, while allowing the world to evolve throughout inference and action execution. Across eight VLMs, agents achieve high task completion yet almost never complete safely: in the hardest setting, only 0.7% of episodes finish without a safety event. More strikingly, matched static and real-time evaluations yield task completion rates of 91.3% and 94.1%, respectively, while real-time execution increases collisions by 12.3×. These results reveal that standard task success can mask substantial safety failures, and that decision latency itself can become a source of physical risk.